Deriv AI Hackathon

Trading

Challenge Brief

Intelligent Trading Analyst

The Challenge

How might we use AI to help traders understand markets, analyze price movements, develop sustainable trading habits, and share insights through engaging social media content?

The Problem

Traders face two interconnected challenges: understanding what's happening in markets and managing their own behavior. Professional traders have analyst teams for market insights and coaches for discipline. Retail traders have neither.

On market understanding:

> *"I see the price dropped 5% but I don't know why. By the time I find the news, the move is over."*
>
> *"There's so much information: economic calendars, news, social media, technical indicators. I don't know what matters."*
>
> *"Professional traders have Bloomberg terminals and analyst teams. I have Google and hope."*

On trading behavior:

> *"I didn't realize I was on a losing streak until I'd lost half my account. No one warned me."*
>
> *"Looking back, I always trade badly when I'm trying to recover losses. I wish I'd seen that pattern earlier."*
>
> *"I know I shouldn't trade when I'm emotional, but in the moment I don't realize I'm being emotional."*

On staying informed:

> *"I follow trading accounts on X and LinkedIn but they're either too basic or trying to sell me something."*
>
> *"I wish there was a trusted voice that explained what's happening in markets without the hype."*
>
> *"I'd love to share market insights with my network, but I don't have time to write quality content."*

The core issue is that trading platforms provide execution but not intelligence. They don't help traders understand what's happening in markets, recognize their own behavioral patterns, or stay connected to a community of informed traders.

Why This Matters Now

Deriv's mission is to make trading accessible to everyone. But accessibility means more than platform access. It means having the insights to understand markets, the self-awareness to trade sustainably, and the content to engage with the trading community.

Traders who understand markets make better decisions. Traders who recognize their behavioral patterns avoid costly mistakes. Traders who engage with quality content stay informed and connected.

Generative AI can now combine market analysis with behavioral insights and content creation, providing comprehensive trading intelligence that was previously available only to professionals.

The Opportunity

Build an AI-powered trading analyst that combines market intelligence, behavioral coaching, and social content generation:

Market Analysis:
- Explain significant price movements in real time ("Why did EUR/USD just spike?")
- Identify technical patterns and explain their significance in plain language
- Summarize relevant news and events affecting specific instruments
- Provide market sentiment analysis from multiple sources
- Generate personalized market briefs for instruments the trader follows

Behavioral Insights:
- Detect patterns indicating emotional or impulsive trading
- Provide gentle, timely nudges when behavior suggests poor decision-making
- Help traders recognize their own winning and losing patterns
- Suggest breaks, limits, or reflection when appropriate
- Celebrate sustainable trading habits, not just profits

Social Media Personas & Content:
- Generate AI analyst personas that share market updates on LinkedIn and X
- Create platform-appropriate content: professional insights for LinkedIn, concise updates for X
- Transform complex market analysis into engaging, shareable posts
- Produce daily/weekly market summaries, notable moves, and educational threads
- Maintain consistent voice and personality across personas
- Generate content calendars with timely, relevant market commentary

The magic is in combining all three: understanding markets, coaching traders, and creating content that builds community and trust.

Constraints

Constraint	Rationale
Must demo live	No slide decks. Show us working software.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
No predictions or signals	The tool analyzes and explains. It does not predict prices or provide buy/sell signals.
Supportive, not restrictive	The system advises and informs. It doesn't block trades or make decisions for traders.
Brand-safe content	Social media content must be compliant, professional, and appropriate for public posting.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does your solution understand both market dynamics and trader psychology?
Usefulness	25%	Would traders rely on this daily? Is it helpful without being patronizing?
Craft	20%	Is the UX considered? Is the AI integration thoughtful?

Ambition	15%	Did you take a risk? Try something unexpected?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

On market analysis:

- What's the difference between information and actionable insight?
- How do you explain technical analysis without being overwhelming?
- How do you filter signal from noise in market news?
- What's the right latency for market analysis to still be useful?

On behavioral insights:

- How do you intervene without being annoying or preachy?
- What's the difference between a bad trade and a bad pattern?
- Can AI detect emotional state from trading behavior alone?
- When is the right moment to suggest a break?

On social media content:

- What makes trading content engaging vs. boring or salesy?
- How do you adapt the same insight for LinkedIn vs. X?
- What personality should an AI analyst persona have?
- How do you build trust and credibility through social content?
- Can personas develop a following that drives platform awareness?

On combining all three:

- How do market conditions affect trader psychology?
- Can the best insights become shareable social content automatically?
- How do you create a cohesive experience across personal insights and public content?

What Would Blow Our Minds

- Real-time market explanations combined with behavioral awareness
- An analyst that says "The market just did X, and based on your history, you tend to Y in these situations"
- AI personas that build genuine followings on LinkedIn and X
- A content engine that turns market analysis into viral threads
- Something that helps traders see both market patterns and their own patterns
- Social content so good that people don't realize it's AI-generated
- A system that catches revenge trading before it starts by connecting market moves to behavior
- Personas with distinct voices: the calm analyst, the data nerd, the trading coach

Something that makes Deriv a trusted voice in trading content

Personalized Trading Education

The Challenge

How might we use Generative AI to provide personalized trading education and strategy explanations tailored to each trader's level and interests?

The Problem

Trading education is usually one-size-fits-all. Beginners get overwhelmed by advanced content. Experienced traders waste time on basics. Nobody gets explanations that match how they actually trade.

> *"I watched 20 YouTube videos on technical analysis. Half were too basic, half were too advanced. None matched what I actually trade."*
>
> *"I lost money on a trade and still don't understand what went wrong. I wish someone would explain it using my actual trade."*
>
> *"Every trading course assumes I want to trade forex. I trade synthetics. The examples never apply to me."*
>
> *"I learn better with examples than theory. But all the educational content is theoretical."*

The core issue is that trading education treats all learners the same. Real learning happens when content adapts to the learner's context, level, and learning style.

Why This Matters Now

Educated traders make better decisions, have better outcomes, and stay longer. But creating personalized educational content at scale was impossible until now.

Generative AI can create customized explanations, generate relevant examples, and adapt teaching style based on the individual learner.

The Opportunity

Build an AI-powered education system that:

- Assesses what the trader already knows and what they need to learn
- Explains concepts using examples from products they actually trade
- Breaks down their own past trades to teach lessons
- Adapts explanation style to how they learn best
- Creates a personalized curriculum based on their goals

Constraints

Constraint	Rationale
Must demo live	No slide decks. Show us working software.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Educational focus	Content teaches concepts and analysis, not "how to win." No promises of profits.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does your solution understand how people actually learn to trade?
Usefulness	25%	Would a trader choose this over generic YouTube tutorials?
Craft	20%	Is the UX considered? Is the AI integration thoughtful?
Ambition	15%	Did you take a risk? Try something unexpected?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

- What's the difference between information and education?
- How do you know if someone actually learned something?
- Can AI explain a trader's own losing trade without making them defensive?
- What makes a trading example memorable vs. forgettable?
- How do you keep education engaging, not boring?

What Would Blow Our Minds

- An AI tutor that explains my trades better than I understand them myself
- Something that creates a curriculum specifically for how I trade
- Education that uses my own wins and losses as teaching moments
- Something that makes learning to trade actually fun

Deriv AI Hackathon

Compliance

Challenge Brief

AI-Powered Transaction Monitoring & Financial Crime Detection

The Challenge

How might we build an AI system that detects money laundering, fraud, and suspicious trading patterns in real-time while dramatically reducing false positives?

The Problem

Financial crime teams are drowning in alerts. Rule-based systems flag thousands of transactions daily. Analysts manually review each one, pulling context from 5+ different systems, documenting decisions, and closing 95% as false positives. Real criminals slip through while analysts burn out.

> *"We get 2,000 alerts per week. I spend my day clicking through accounts, checking if that $5,000 deposit is suspicious. Most aren't. The real fraud hides in the noise."*
>
> *"Client deposits $500, trades for 10 minutes with $0.01 profit, then withdraws. That's not trading - that's laundering. But our rules didn't catch it because the amounts were 'normal.'"*
>
> *"By the time we reconstruct a fraud timeline manually - pulling trades, deposits, IP logs, device IDs - the money is gone and the account is abandoned."*

The core issue is high-volume noise from rigid rules, slow manual investigation, and reactive detection that's always one step behind sophisticated criminals.

Why This Matters Now

Deriv processes millions of transactions monthly across deposits, trades, and withdrawals. Financial crime risks include:

- Money laundering: Fake trades, rapid deposit-withdrawal cycles, layering through multiple accounts
- Payment fraud: Stolen cards, chargebacks, coordinated fraud rings
- Market abuse: Opposite trading, insider trading, manipulation attempts
- Regulatory requirements: Must file SARs/STRs for suspicious activity within tight deadlines

Current rule-based systems generate alert fatigue. The compliance team needs AI that finds needles in haystacks and explains why something is suspicious.

The Opportunity

Build an AI-powered transaction monitoring system that:

- Behavioral anomaly detection: ML models that learn normal patterns per customer and flag deviations
- Network analysis: Graph algorithms that identify coordinated fraud rings (shared IPs, devices, cards, timing patterns)
- Contextual scoring: Risk scores that consider transaction history, trading behavior, deposit/withdrawal patterns, KYC data

- Automated evidence collection: When flagged, automatically pulls relevant logs, trades, communications, and timeline
- False positive reduction: Learns from analyst decisions to reduce noise over time
- Real-time intervention: Flags suspicious withdrawals before funds leave, not 3 days later
- SAR preparation: Auto-generates suspicious activity report drafts with evidence and narrative

The system should turn 2,000 alerts into 50 high-confidence cases with full investigation packs ready for review.

Constraints

Constraint	Rationale
Must demo live	No slide decks. Show us working software.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Minimize false positives	Alert fatigue damages analyst effectiveness. Noise kills productivity.
Explainable decisions	Analysts and regulators need to understand why activity is flagged.
Real-time processing	Detection must happen during transactions, not days later.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does your solution understand the real challenges of financial crime detection?
Usefulness	25%	Would compliance analysts actually use this? Does it save real investigation time?
Craft	20%	Is the UX considered? Is the AI integration thoughtful?
Ambition	15%	Did you take a risk? Try something unexpected?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

- How do you balance catching sophisticated fraud vs. not flagging legitimate high-volume traders?
- What graph analysis techniques best reveal coordinated fraud networks?
- Should the system prioritize certain fraud types (laundering vs. card fraud vs. trading abuse)?
- How do you handle the cold-start problem with new customers who have no behavioral history?
- Can you generate regulator-ready narratives explaining suspicious activity?
- What makes an alert actionable vs. just noise?
- How do you explain complex network relationships to non-technical compliance officers?

What Would Blow Our Minds

- Unsupervised learning that discovers entirely new fraud typologies not in training data
- Temporal pattern recognition: "This account's behavior changed dramatically 72 hours after KYC approval"

- Network effect analysis: "This fraud ring has 47 linked accounts we didn't know about"
- Predictive flagging: Score accounts as high-risk before they commit fraud based on early signals
- Automated SAR generation that meets regulatory quality standards with minimal human editing
- Something that turns 2,000 weekly alerts into 50 high-confidence cases
- A system that catches the $0.01 profit laundering that rules completely miss

AI Compliance Manager

The Challenge

How might we build an AI compliance manager that continuously monitors customer behavior patterns, automatically detects activities inconsistent with their declared profile, and stays current with evolving regulatory requirements?

The Problem

Compliance teams are drowning in data. Thousands of customers transact daily, but only a handful exhibit truly suspicious behavior. Manual monitoring catches obvious cases too late, misses subtle patterns, and generates false positives that waste analyst time.

> *"We know regulators expect us to spot transactions inconsistent with customer profiles. But how do I review 10,000 accounts for behavioral anomalies? I can't."*
>
> *"A customer's behavior changes dramatically overnight - deposit patterns, transaction sizes, frequency. By the time we notice manually, weeks have passed."*
>
> *"We set rigid rules: flag deposits over $10,000. Great - now I have 500 alerts per day, 95% are false positives, and the real criminals slip through with $9,999 deposits."*
>
> *"Different analysts interpret 'suspicious' differently. What one person flags, another approves. We need consistency."*

And then there's the regulatory landscape:

> *"A new FATF recommendation dropped last month. I still don't know how it affects our monitoring rules."*
>
> *"We operate in 15 jurisdictions. Each has different requirements. Keeping track of what changed where is a full-time job."*
>
> *"The regulator updated their guidance on crypto transactions. By the time legal translated it into policy changes, we were already three months behind."*
>
> *"I found out about a regulatory change from a competitor's press release. That's not how compliance should work."*

The core issue: human-scale monitoring can't match machine-scale activity, and the regulatory landscape changes faster than teams can adapt. Sophisticated risks hide in volume, reactive compliance is always too late, and regulatory changes create blind spots.

Why This Matters Now

Financial regulators globally require institutions to identify unusual customer behavior, but they don't prescribe exactly how. Modern financial crime is sophisticated:

- Criminals know common thresholds and stay just below them
- Behavior patterns are more telling than single transactions
- Context matters: what's normal for one customer is suspicious for another
- Early detection prevents losses and regulatory breaches

Meanwhile, the regulatory environment is constantly evolving:

- New AML/CFT requirements emerge from FATF, regional bodies, and local regulators

- Guidance on emerging risks (crypto, digital payments, AI-enabled fraud) updates frequently
- Enforcement actions against other firms signal changing regulatory expectations
- Operating across multiple jurisdictions multiplies the complexity

Compliance teams need AI that thinks like an experienced analyst but operates at machine scale - spotting patterns humans miss, learning from feedback, explaining its reasoning, and staying current with regulatory changes that could affect monitoring rules.

The Opportunity

Build an AI-powered compliance manager that:

Behavioral Monitoring:

- Behavioral profiling: Learns what's "normal" for each customer based on their profile (income, employment, location) and historical behavior
- Anomaly detection: Flags deviations from established patterns - sudden spikes in activity, unusual transaction sizes, geographic inconsistencies
- Contextual intelligence: Considers multiple signals together rather than evaluating transactions in isolation
- Risk scoring: Assigns confidence scores to anomalies to prioritize analyst attention
- Adaptive learning: Improves detection accuracy based on analyst feedback (true vs. false positives)

Regulatory Intelligence:

- Regulatory monitoring: Tracks changes from relevant regulators, standard-setters (FATF), and enforcement bodies across all operating jurisdictions
- Impact analysis: Automatically assesses how regulatory changes affect current monitoring rules and customer populations
- Gap identification: Highlights where current processes may not meet new or updated requirements
- Policy recommendations: Suggests specific changes to thresholds, rules, or procedures based on regulatory updates
- Compliance calendar: Tracks implementation deadlines and provides proactive alerts

Operational Integration:

- Automated workflows: Triggers appropriate responses based on risk level (alert analyst, request additional verification, apply temporary restrictions)
- Audit trail: Documents all detections, decisions, and reasoning for regulatory inspection
- Regulatory reporting: Helps prepare required reports and filings with pre-populated data

The system should reduce false positives by 80% while catching 95%+ of genuine compliance risks, explain its reasoning clearly, and ensure the team is never caught off-guard by regulatory changes.

Constraints

Constraint	Rationale
Must demo live	No slide decks. Show us working software.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Explainable decisions	Compliance officers and regulators need to understand why something was flagged.

Minimize false positives	Alert fatigue reduces effectiveness and frustrates legitimate customers.
Handle diverse profiles	From high-net-worth professionals to students with part-time jobs.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does your solution understand what's "normal" for each specific customer?
Usefulness	25%	Would compliance officers actually use this? Does it reduce workload?
Craft	20%	Is the UX considered? Is the AI integration thoughtful?
Ambition	15%	Did you take a risk? Try something unexpected?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

On behavioral monitoring:

- How do you define "normal" for a customer with only 2 weeks of history?
- What's the right balance between being too sensitive (many false alarms) vs. too lenient (missing real risks)?
- How do you weight different signals - is a geographic anomaly more suspicious than a transaction size anomaly?
- How do you handle legitimate life changes (job promotion, inheritance) vs. suspicious behavior?

On regulatory intelligence:

- How do you determine which regulatory changes are relevant to Deriv's operations?
- Can AI accurately interpret regulatory language and translate it into monitoring rule changes?
- How do you prioritize regulatory changes - some are urgent, others have long implementation timelines?
- How do you handle conflicting requirements across different jurisdictions?
- Can the system predict regulatory trends based on enforcement patterns and consultation papers?

What Would Blow Our Minds

- Unsupervised learning that discovers entirely new risk patterns not defined in rules
- Temporal intelligence: "This behavior was fine last month, but given their recent activity changes, it's now suspicious"
- Network effects: "This customer's behavior matches a fraud ring we detected in 3 other accounts"
- Predictive flagging: Early warning signals before compliance breaches occur
- Natural language explanations: "Flagged because: deposits increased 400% while declared income unchanged, similar to 12 confirmed money laundering cases"
- Regulatory radar: "New FATF guidance on virtual assets published yesterday - here's how it affects 3 of your current monitoring rules"
- Impact simulation: "This regulatory change will increase alerts by 15% in the Malta jurisdiction - here's a recommended threshold adjustment"
- Compliance gap alerts: "Based on recent FCA enforcement actions, your current approach to transaction monitoring may be below emerging expectations"
- Something that ensures we're never surprised by a regulatory change again

Deriv AI Hackathon

Anti-Fraud

Challenge Brief

AI-Powered Client Fraud Detection & Investigation

The Challenge

How might we build an AI system that detects, investigates, and explains suspicious client activity - from identity fraud to money laundering - while reducing false positives and empowering investigators to work faster and smarter?

The Problem

Fraud teams face an impossible task: review thousands of alerts daily, investigate suspicious accounts across multiple systems, distinguish real fraud from false positives, and document everything for regulators - all while fraudsters evolve faster than rule-based systems can adapt.

> *"I review 200 alerts per day. Maybe 10 are real fraud. But I can't skip any because the one I skip might be the critical one."*
>
> *"The system flags an account as 'high risk' with a score of 87. But it doesn't tell me WHY. I spend an hour gathering data from five different systems just to understand what triggered it."*
>
> *"By the time I investigate one case, ten more pile up. And half of them are false positives that look identical to real fraud."*
>
> *"We discovered a new fraud pattern last month. Looking back, it had been happening for six months - we just didn't see it until losses got big enough to notice."*
>
> *"A student declares $500 monthly income but deposits $8,000 in two weeks. An account uses AI-generated faces for verification. Someone accesses from a sanctioned country via VPN. These all require investigation, but I don't have enough hours in the day."*

The core issue: investigators spend 80% of their time on data gathering and noise, 20% on actual fraud detection and decision-making. Meanwhile, sophisticated fraud hides in plain sight.

Why This Matters Now

Financial platforms face increasingly sophisticated client-side fraud:

- Identity fraud: Synthetic IDs, deepfakes, stolen documents, AI-generated faces
- Money laundering: Fake trades, rapid deposit-withdrawal cycles, transactions inconsistent with customer profiles
- Account takeover: Credential stuffing, phishing, social engineering
- Behavioral anomalies: Deposits far exceeding declared income, unusual geographic access patterns
- Sanctions evasion: VPN usage to mask location, accessing from prohibited jurisdictions

Current challenges:

- Alert overload: Rule-based systems generate 95%+ false positives
- Manual investigation: Data gathering takes hours per case

- Reactive detection: Fraud patterns discovered months after they start
- Inconsistent decisions: Different investigators interpret the same signals differently
- Documentation burden: Writing investigation reports is time-consuming
- Evolving threats: Fraudsters adapt faster than static rules can be updated

Fraud teams need AI that detects, explains, prioritizes, investigates, and learns - continuously.

The Opportunity

Build an integrated AI fraud detection and investigation system with four core capabilities:

1. Intelligent Detection & Pattern Discovery:
   - Multi-signal detection: Monitor identity verification, transaction behavior, geographic access, and account activity
   - Behavioral anomaly detection: Flag deposits inconsistent with declared income, unusual transaction patterns, geographic impossibilities
   - Identity fraud detection: Detect forged documents, synthetic IDs, AI-generated faces, deepfakes, biometric mismatches
   - Sanctions & geographic risk: Identify VPN usage, access from prohibited jurisdictions, location anomalies
   - Emerging pattern discovery: Use unsupervised learning to discover new fraud typologies before they become widespread
   - Network analysis: Connect related accounts through shared IPs, devices, timing patterns, or coordinated behavior

2. Explainable Alerts & Prioritization:
   - Clear explanations: Generate human-readable explanations - "Flagged because: deposits 1400% above declared income + 73% of logins from high-risk jurisdiction via VPN"
   - Confidence scoring: Risk levels with reasoning - why is this 87 vs. 65?
   - Intelligent prioritization: Learn from historical outcomes to rank alerts by likelihood of being real fraud
   - False positive prediction: Identify and auto-resolve obvious false positives with audit trail
   - Contextual comparison: "This pattern matches 12 confirmed money laundering cases from last quarter"

3. AI-Assisted Investigation:
   - Automated data gathering: Pull relevant information from multiple systems into a coherent case view in seconds
   - Timeline reconstruction: Build chronological sequences of suspicious activity automatically
   - Evidence synthesis: Summarize transaction history, login patterns, document verification results, and behavioral signals
   - Investigation suggestions: Recommend next steps - "Check if this IP is shared with other accounts"
   - Cross-case connections: "This device was used by 7 other flagged accounts"

4. Documentation & Learning:
   - Automated report generation: Create investigation summaries and audit-ready compliance documentation
   - Regulatory explanations: Generate findings suitable for different audiences (investigators, compliance, regulators)
   - Continuous learning: Improve detection accuracy based on investigator feedback and confirmed outcomes

- Knowledge capture: Learn from every investigation to improve future detection

The system should turn a 2-hour investigation into a 10-minute review while catching fraud that manual processes miss.

Constraints

Constraint	Rationale
Must demo live	Show working software, not concept slides.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Human in the loop	AI assists and recommends; investigators make final decisions.
Explainable decisions	Every alert and recommendation must have clear reasoning.
No missed fraud	False positive reduction cannot sacrifice fraud detection.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does it catch real fraud? Discover new patterns? Reduce false positives?
Usefulness	25%	Does it save real time? Would investigators use it daily?
Craft	20%	Are explanations clear, accurate, and actionable? Is the UX intuitive?
Ambition	15%	Did you take risks? Try something unexpected?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

- What's the most time-consuming part of fraud investigation that AI could accelerate?
- How do you balance comprehensive detection with manageable alert volumes?
- What makes an explanation actionable vs. just informative?
- How do you distinguish emerging fraud patterns from random anomalies or legitimate business changes?
- When should AI suggest next steps vs. wait to be asked?
- How do you handle cases where multiple fraud signals conflict?
- Can you connect dots across identity, transactions, and behavior that investigators wouldn't see?
- How do you present complex case information clearly without overwhelming the investigator?

What Would Blow Our Minds

- Investigation copilot: Complete case summary with evidence, timeline, and recommendations in 30 seconds
- Pattern discovery: Finding fraud techniques weeks before they become obvious - "Detected emerging behavior cluster: 47 accounts with similar transaction patterns"
- Network revelation: "This account is part of a 23-account fraud ring sharing 3 devices and coordinating transaction timing"
- Predictive explanations: "This alert is 89% likely to be real fraud based on 5 strong signals matching confirmed cases"
- Learning from outcomes: System that improves daily based on investigator decisions and discovered patterns

- Cross-domain intelligence: "Identity documents passed automated checks, but transaction behavior + VPN usage + device fingerprint all match known fraud network"
- Audit-ready documentation: Investigation reports generated automatically that meet regulatory standards
- 5x productivity: Investigators handling 5x more cases with higher accuracy

AI-Powered Partner & Affiliate Fraud Detection

The Challenge

How might we build an AI system that detects fraudulent behavior by partners and affiliates - from opposite trading schemes to fake traffic to commission abuse - while distinguishing sophisticated fraud from legitimate business activity?

The Problem

Partner and affiliate fraud is different from client fraud. It's not about stolen credit cards or fake IDs - it's about exploiting business relationships and commission structures. Fraudulent partners are often sophisticated, understand the rules, and know how to stay just under detection thresholds.

> *"We have 5,000 active affiliates. A handful are running coordinated opposite trading schemes across dozens of referred accounts. But they spread it out so it's hard to see the pattern."*
>
> *"Partner refers 100 'clients' who all deposit, make one tiny trade, then withdraw. That's not real business - that's bonus abuse. But proving it requires manual investigation of each account."*
>
> *"Two affiliates refer clients who consistently take opposite positions with identical timing. They're splitting profits. But they're not explicitly linked in our system."*
>
> *"By the time we notice the pattern - affiliate refers clients who all behave suspiciously - they've already collected months of commissions."*
>
> *"Our SO (sub-affiliate) compensation structure is being gamed. We see earnings patterns that don't match legitimate business, but investigating takes weeks of pulling data from multiple systems."*

The core issue: partner fraud is relational and pattern-based, not transactional. It requires analyzing networks of accounts, timing correlations, and business behavior patterns that manual reviews can't scale to detect.

Why This Matters Now

Trading platforms rely on partner and affiliate networks for customer acquisition, but these relationships create fraud opportunities:

- Opposite trading: Partners coordinate referred clients to take opposite positions and split profits
- Mirror trading: Affiliated accounts trade in lockstep across different affiliates
- Bonus abuse: Fake sign-ups designed to harvest referral bonuses without real trading activity
- Commission fraud: Manipulating metrics to inflate affiliate payouts
- Client recycling: Same individuals signing up through multiple affiliates to game promotions
- Traffic fraud: Fake clicks, bot traffic, or stolen leads to inflate performance

Current challenges:

- Relationship complexity: One partner may control dozens of referred clients across multiple levels (sub-affiliates)
- Delayed detection: Patterns become obvious only after months of activity
- Sophisticated evasion: Fraudulent partners understand detection systems and work around them
- Investigation burden: Mapping affiliate networks and analyzing correlated behavior takes days per case
- Attribution difficulty: Proving which partner is responsible when fraud spans multiple levels

Partner teams need AI that understands business relationships, detects behavioral coordination, and maps fraud networks.

The Opportunity

Build an AI-powered partner fraud detection system with four core capabilities:

1. Network-Based Fraud Detection:
    - Relationship mapping: Visualize partner hierarchies, referred client networks, and sub-affiliate structures
    - Opposite trading detection: Identify pairs or groups of accounts (across different affiliates) taking opposite positions with correlated timing
    - Mirror trading patterns: Detect accounts that trade in lockstep - same instruments, similar timing, coordinated entry/exit
    - Client clustering: Find groups of referred clients with suspiciously similar behavior (deposit patterns, trading style, geographic signals)
    - Cross-affiliate analysis: Detect when the same individual appears as a client under multiple affiliate IDs
    - Temporal correlation: Spot coordinated activity timing across ostensibly unrelated accounts

2. Business Behavior Analysis:
    - Fake traffic detection: Identify bot-driven sign-ups, stolen leads, or artificial traffic inflation
    - Bonus abuse patterns: Flag referrals designed to harvest bonuses without genuine trading (minimal trades, immediate withdrawals)
    - Commission anomalies: Detect earnings inconsistent with legitimate business metrics (volume, client retention, trading activity)
    - Client quality scoring: Assess whether referred clients exhibit real trading behavior or orchestrated activity
    - Performance manipulation: Identify partners gaming metrics to inflate payouts
    - Lifecycle analysis: Track referral-to-active-trader conversion patterns and flag statistical outliers

3. Explainable Investigation Support:
    - Fraud hypotheses: "Partner A refers 47 clients; 23 consistently take opposite positions to clients referred by Partner B; timing correlation: 89%"
    - Evidence synthesis: Automatically gather transaction data, affiliate earnings, client behavior, and network relationships
    - Timeline visualization: Show evolution of suspicious patterns over weeks or months
    - Commission impact calculation: Quantify financial exposure from suspected fraud
    - Confidence scoring: "High confidence: 5 of 5 indicators match known opposite trading scheme"
    - Investigation suggestions: "Check if these partners share IP, payment methods, or KYC documents"

4. Proactive Pattern Discovery & Learning:
- Emerging scheme detection: Discover new fraud techniques before they scale
- Fraud ring identification: Use graph analysis to map coordinated fraud networks spanning multiple affiliate levels
- Anomaly surfacing: Flag statistical outliers worth investigating before losses accumulate
- Continuous learning: Improve detection based on confirmed fraud cases and investigator feedback
- Predictive alerting: "Partner X's recent referral patterns match pre-fraud signatures seen in 8 historical cases"

The system should turn a multi-week partner fraud investigation into a same-day analysis with clear evidence trails.

Constraints

Constraint	Rationale
Must demo live	Show working software, not concept slides.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Relationship-aware	Must understand partner hierarchies and attribution chains.
Evidence-based	Accusations require clear, defensible evidence.
Human in the loop	AI detects and recommends; partner managers make final decisions.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does it detect coordinated fraud across partner relationships?
Usefulness	25%	Does it find opposite trading, mirror trading, and novel schemes?
Craft	20%	Does it save partner managers real time? Are fraud allegations backed by clear evidence?
Ambition	15%	Did you take risks? Try something unexpected?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

- How do you distinguish coordinated fraud from coincidence in trading patterns?
- What statistical thresholds separate suspicious correlation from legitimate business?
- How do you visualize complex partner networks in an intuitive way?
- Can you detect fraud that spans multiple levels of sub-affiliates?
- How do you handle new fraud techniques the system hasn't seen before?
- What makes evidence convincing enough to justify partner suspension?
- How do you balance early detection with avoiding false accusations?
- Can you quantify the financial impact of detected fraud schemes?

What Would Blow Our Minds

- Fraud ring mapping: "Detected coordinated network: 7 partners, 23 sub-affiliates, 184 referred clients engaged in opposite trading scheme; estimated fraud value: $X"
- Temporal intelligence: "Partner behavior changed 6 weeks ago; current pattern matches fraud signatures; predicted escalation in next 2 weeks"

- Emerging pattern discovery: "New scheme detected: 12 partners with statistically impossible client conversion rates and identical trading behavior distribution"
- Cross-level attribution: "Root cause: Master Partner A; executed through 3 layers of sub-affiliates to obscure connection"
- Evidence automation: Complete investigation package with relationship maps, transaction timelines, statistical analysis, and commission calculations - ready in minutes
- Proactive prevention: "Warning: Partner X's recent referral patterns match pre-fraud signatures; recommend enhanced monitoring"

  Learning system: Improves detection based on confirmed cases and discovers entirely new fraud typologies through unsupervised learning

Deriv AI Hackathon

Finance

Challenge Brief

AI-Powered Payments Approval & Fraud Intelligence

The Challenge

How might we build an AI system that automates payment approvals and detects transaction fraud in real-time - turning a manual bottleneck into instant, intelligent decisions?

The Problem

Payments teams review hundreds of payout requests daily, checking multiple systems for each one. Meanwhile, fraud hides in plain sight because manual reviews can't keep pace with sophisticated patterns.

> *"I approve payouts all day. Each one requires checking multiple systems for red flags. Most could be instant, but the system can't decide - so I click through the same checks over and over."*
>
> *"Client uses a restricted card. I see the error a day later, manually review the account, then lock it. By then, they've made more transactions."*
>
> *"Fraudster deposits, makes one tiny trade, withdraws immediately. I know the pattern, but I'm reviewing these cases one by one after they've already happened."*
>
> *"When a fraud incident hits - stolen cards, coordinated abuse - I prepare files to lock accounts, recover funds, notify clients. Takes hours. High error risk."*

The core issue: manual approval doesn't scale, and reactive fraud detection is always too late. AI can approve the legitimate and flag the suspicious - instantly.

Why This Matters Now

Trading platforms process thousands of daily payments across dozens of payment methods. The industry faces common challenges:

- Low auto-approval rates - payments officers spending hours on routine decisions that could be automated
- Fraud detected too late - card errors, suspicious patterns found after losses occur
- No real-time risk scoring - decisions based on incomplete information
- Slow incident response - bulk actions require manual preparation
- Alert fatigue - teams buried in false positives while real fraud slips through

The result: legitimate customers wait unnecessarily, fraudsters exploit detection gaps, and teams drown in reactive work.

The Opportunity

Build an AI-powered payments intelligence system that makes two critical decisions faster and better than humans:

1. Intelligent Payout Approval:
   - Instant risk assessment: Evaluate every payout request against multiple fraud signals in seconds
   - Multi-signal decisioning: Payment method status, transaction velocity, geographic consistency, trading behavior, historical patterns

- Smart auto-approval: Process the vast majority of clean payouts instantly with clear reasoning
- Intelligent escalation: Flag suspicious cases with evidence for human review
- Continuous learning: Improve accuracy based on officer feedback

2. Real-Time Fraud Detection & Response:

- Behavioral pattern detection: "No trade" fraud (deposit → minimal trading → immediate withdrawal), "short trade" abuse, card testing, velocity abuse
- Technical fraud signals: Card errors, failed login attempts, geographic impossibilities, device fingerprint anomalies
- Automated response: High risk → auto-lock + notify; Medium risk → flag + monitor; Low risk → approve + log
- Bulk incident handling: One-click bulk locks, automated fund recovery, auto-generated client communications
- Real-time exposure calculation: Instant quantification of fraud ring impact

Example - Auto-Approved Payout:
Withdrawal request → AUTO-APPROVED (High Confidence)

- Payment method verified (used multiple times, no issues)
- Amount within normal range for this customer
- Geographic match (consistent with recent logins)
- Active trading history
- Processing time: seconds

Example - Escalated for Review:
Withdrawal request → ESCALATED (Elevated Risk Score)

- New payment method (first time use)
- Unusual amount (significantly higher than previous withdrawals)
- Geographic anomaly (VPN detected)
- Minimal trading activity
- Pattern match: Similar to confirmed fraud cases

Example - Fraud Detection:
FRAUD ALERT: "No Trade" Fraud Pattern Detected (High Confidence)

- Recent deposit via card
- Minimal trading activity (single small trade, closed immediately)
- Withdrawal request for nearly full deposit amount
- Automated actions: Withdrawal blocked, account flagged, fraud team notified, related accounts identified

The system should turn manual approval bottlenecks into instant, intelligent decisions while catching fraud in real-time.

Constraints

Constraint	Rationale
Must demo live	Show real automation with sample transactions.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Explainable decisions	Every approval/block must have clear reasoning.
Real-time performance	Decisions in seconds, not minutes or hours.
Human oversight	AI decides routine cases; humans handle edge cases.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does it correctly approve clean payouts and flag fraud?
Usefulness	25%	Can it dramatically increase auto-approval rates? Detect fraud in real-time?
Craft	20%	Are decisions clear enough for officers to trust and defend?
Ambition	15%	Creative approaches to pattern detection or decision logic?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

- What signals are most predictive of fraud vs. false alarms?
- How do you handle brand-new customers with no behavioral history?
- When should AI be conservative (block) vs. aggressive (approve)?
- Can you detect fraud patterns that don't match any known typology?
- How do you balance speed (instant approval) with accuracy (catch fraud)?
- What's the right interface for fraud investigators reviewing alerts?
- Can the system explain why one withdrawal is approved and another flagged?

What Would Blow Our Minds

- Dramatic auto-approval increase: Safely processing the vast majority of payouts instantly while maintaining accuracy
- Predictive fraud flagging: Identifying accounts showing pre-fraud signatures before they attempt withdrawal
- Real-time pattern discovery: Detecting new fraud patterns and proactively flagging similar accounts
- Incident response in seconds: Detecting fraud rings and executing bulk containment actions instantly
- Learning from outcomes: Measurable improvement in approval confidence based on validated decisions
- Natural language fraud queries: "Show me accounts that deposited but traded minimally"

Fraud network visualization: Graph showing connected accounts via shared devices/IPs/cards

AI-Powered Financial Reconciliation Assistant

The Challenge

How might we build an AI system that automates daily financial reconciliation across multiple payment providers - turning a manual, error-prone process into instant, accurate matching?

The Problem

Finance teams manually extract data from multiple payment service providers (PSPs) every day, then spend hours reconciling what PSPs report vs. what's in internal systems vs. what's in the ERP. Discrepancies take days to investigate. By month-end, it's chaos.

> *"Every morning: download transaction files from payment providers. Load into spreadsheets. Compare PSP balances to internal balances to ERP records. Takes hours before I can even start investigating discrepancies."*
>
> *"Found a variance between a PSP and our internal system. Now I manually hunt through transactions: Is it a missing deposit? FX rate mismatch? Duplicate entry? Takes hours per discrepancy."*
>
> *"Month-end is a nightmare: reconcile PSP statements to calculated balances, match aggregated client transactions to source systems, verify account balances. All manual. Takes days."*
>
> *"We discover problems too late - missing transactions, FX errors, duplicate entries found at month-end instead of daily. Then scramble to fix before closing books."*

The core issue: reconciliation is 90% data extraction and matching, 10% investigation. AI can automate the 90% so humans focus on the 10% that matters.

Why This Matters Now

Financial platforms process millions of transactions monthly across multiple PSPs, currencies, and entities. The industry faces common challenges:

- Manual PSP data extractions daily - consuming hours before analysis even begins
- Delayed discrepancy detection - problems found at month-end, not daily
- Manual investigation - every variance requires hunting through multiple systems
- No real-time visibility - working with yesterday's data to reconcile last week's transactions
- High error risk - manual data entry and matching prone to mistakes

The result: Finance spends most of their time on data manipulation instead of actual financial analysis. Can't provide real-time insights. Discovers problems too late.

The Opportunity

Build an AI-powered reconciliation assistant that automates the data work and intelligently resolves discrepancies:

1. Automated Multi-Source Reconciliation:
    - Intelligent data extraction: Automatically pull transactions and balances from multiple PSPs

- Handle different file formats: CSV, JSON, XML, API responses with varying date formats, currencies, transaction types
- Smart matching algorithms: Exact matches, fuzzy matches (amount + date with tolerance for timing differences), pattern matches (batch transactions, split transactions, FX conversions)
- Three-way reconciliation: PSP transactions ↔ Internal client records ↔ ERP accounting
- Variance calculation: Identify which system(s) have discrepancies with amounts and directions

2. Intelligent Discrepancy Resolution:
- Root cause analysis: Missing transactions, timing differences, FX mismatches, duplicate entries, fee discrepancies
- Evidence gathering: Automatically compile transaction IDs, timestamps, FX rates, similar historical cases
- Resolution recommendations: Auto-generate adjustment entries, suggest manual actions, calculate financial impact
- Pattern learning: Remember how similar discrepancies were resolved and suggest same approach

3. Proactive Alerts & Insights:
- Real-time anomaly detection: Unusual balance changes, unexpected patterns, potential PSP issues
- Predictive warnings: Based on historical patterns, anticipate which discrepancies will self-resolve vs. need action
- Trend analysis: Identify recurring issues, track PSP reliability, recommend process improvements
- Automated reporting: Daily summaries, weekly exception reports, monthly audit-ready documentation

Example - Daily Auto-Reconciliation:
Daily Reconciliation Summary:

- AUTO-RECONCILED: Vast majority of PSPs - transactions matched perfectly or auto-adjusted for timing
- INVESTIGATION REQUIRED: Small number of PSPs with variances
- Root cause hypotheses and evidence packages prepared automatically

Example - Discrepancy Analysis:
Discrepancy Detected: PSP variance (PSP higher than internal system)

- Root Cause Identified: Missing deposits in internal system
- Evidence: Specific transactions found in PSP but not in internal records
- Recommended Actions: Create deposit adjustments, credit client accounts, update accounting entries
- Impact Assessment: Priority level assigned based on client impact

The system should dramatically reduce month-end reconciliation time while catching issues daily instead of monthly.

Constraints

Constraint	Rationale
Must demo live	Show actual reconciliation automation with sample data.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Accuracy critical	Financial errors have regulatory and audit consequences.

Explainable matching	Finance teams need to understand why AI matched/didn't match.
Audit trail required	All adjustments must be documented for regulatory compliance.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Can it correctly match transactions across different formats/timing?
Usefulness	25%	What % of reconciliation can be automated vs. requiring human review?
Craft	20%	Does it accurately diagnose WHY discrepancies exist?
Ambition	15%	Creative approaches to fuzzy matching or pattern recognition?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

- What AI techniques work best for matching transactions with timing differences?
- How do you handle transactions that are "close but not exact" (amount off by fees)?
- Can you learn reconciliation patterns from historical data?
- How do you explain complex three-way discrepancies simply?
- What's the right balance between auto-resolving vs. escalating for review?
- Should the system predict WHEN discrepancies will resolve themselves?
- How do you handle many different file formats and data structures?

What Would Blow Our Minds

- High auto-reconciliation rate: Vast majority of PSPs reconciled automatically with minimal escalations
- Intelligent root cause analysis: Not just "amounts don't match" but specific diagnosis with recommended actions
- Pattern learning: Recognizing PSP-specific behaviors (e.g., "this PSP always has weekend lag") and flagging accordingly
- Predictive issue detection: Warning about unusual patterns before they become problems
- One-click resolution: Generated adjustment entries ready for review and execution
- Dramatic time savings: Month-end reconciliation reduced from days to hours

Natural language queries: "Show me all discrepancies from this PSP last month" with instant, accurate results

Deriv AI Hackathon

Partners & Affiliates

Challenge Brief

AI Partner Churn Prediction & Early Warning System

The Challenge

How might we build an AI system that predicts which partners are at risk of leaving, identifies why they're disengaging, and alerts relationship managers before it's too late?

The Problem

Partner churn is invisible until it's irreversible. By the time you notice a partner has stopped sending traffic or is promoting competitors, months of revenue are gone and the relationship is already dead.

> *"Our top partner in the region went from high performer to barely active over a few months. We didn't notice until it was too late."*
>
> *"Partner stopped logging into the dashboard weeks ago. No one flagged it. Now they're promoting a competitor."*
>
> *"We have thousands of active partners. I manage hundreds personally. How do I know which ones need attention this week?"*
>
> *"Partner submitted multiple support tickets about commission discrepancies. No one connected the dots. A few months later, they churned."*

The core issue: relationship managers are reactive, not proactive. They only notice problems when partners explicitly complain or when performance has already collapsed. Silent disengagement goes undetected.

Why This Matters Now

Affiliate and partner networks drive a significant portion of customer acquisition for most platforms. Partner churn is expensive:

- Recruiting costs significantly more than retention
- Revenue loss is delayed: Partners slowly reduce activity before leaving completely
- High-value churn hurts most: A small percentage of top partners often generate the majority of referrals
- Competitive poaching: Rivals actively recruit your best performers
- Scale problem: Can't personally monitor thousands of relationships

Traditional CRM doesn't help because it tracks what happened, not what's about to happen. Managers need predictive intelligence.

The Opportunity

Build an AI-powered partner churn prediction system that:

Early Warning Detection:
- Activity pattern analysis: Track partner engagement trends over time
- Login frequency declining
- Referral volume dropping (week-over-week, month-over-month)
- Traffic quality decreasing
- Commission earnings falling

- Dashboard usage patterns changing
- Engagement degradation scoring: Quantify how partner behavior is shifting compared to their historical baseline

Multi-Signal Risk Scoring:
- Behavioral signals: Activity decline, engagement drop-off, performance degradation
- Communication signals: Support ticket volume/sentiment, response time to outreach, feedback tone
- Relationship signals: Event attendance, training participation, program adoption
- Financial signals: Commission trends, payout issues, payment disputes
- Competitive signals: Mentions of competitors, pricing complaints, feature requests

Churn Probability Prediction:
- Risk scoring: Assign churn probability to each partner (High / Medium / Low risk tiers)
- Explainable predictions: Clear reasoning for risk scores
- Example: "High risk because: Login frequency declining significantly, referral volume down, unresolved support tickets, competitor mentioned in recent feedback"

Intelligent Alerting & Prioritization:
- Real-time alerts: Notify partner managers when risk scores cross thresholds
- Priority ranking: Focus manager attention on highest-value, highest-risk partners
- Segment analysis: Identify when multiple partners in a region or segment show similar churn signals - possible systemic issue

The system should detect disengagement well before churn occurs, with enough lead time for effective intervention.

Constraints

Constraint	Rationale
Must demo live	Show actual churn prediction on sample partner data.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Explainable predictions	Managers need to understand WHY someone is at risk.
Minimize false positives	Alert fatigue damages manager effectiveness.
Actionable insights	Risk scores alone aren't enough - need context for intervention.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does it correctly identify partners who will churn?
Usefulness	25%	Can it predict well in advance with useful lead time for intervention?
Craft	20%	Does it find patterns humans would miss? Is the UX intuitive?
Ambition	15%	Did you take risks? Try something unexpected?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

- What behavioral signals are most predictive of churn?
- How do you distinguish temporary performance dips from genuine disengagement?
- What's the right prediction window for useful intervention?
- Should alerts be partner-specific or identify cohort-level trends?
- How do you balance sensitivity (catch all churn) vs. specificity (avoid false alarms)?

- Can the system learn which interventions successfully prevent churn?

What Would Blow Our Minds

- Accurate early prediction: High churn probability identified with clear reasoning based on multiple declining signals
- Cohort risk detection: "Warning: Multiple partners in this region showing similar churn signals - investigate systemic issue"
- Intervention timing: Identifying the optimal intervention window before churn probability becomes too high
- Pattern discovery: "New churn pattern detected: partners with unresolved technical issues for extended periods have significantly higher churn rates"
- Learning from outcomes: System that improves predictions based on which partners actually churned vs. were successfully retained
- Proactive recommendations: Not just alerting to risk, but suggesting specific retention actions based on the churn drivers identified

Deriv AI Hackathon

HR

Challenge Brief

Self-Service HR Operations Platform

The Challenge

How might we build an AI-powered self-service platform where contracts generate themselves, queries answer themselves, and HR operations become invisible?

The Problem

HR teams drown in repetitive operational tasks that consume the majority of their time:

> *"Every hire needs an offer letter, employment contract, NDA, equity docs - each customized manually. It takes hours per person."*
>
> *"Employees ask the same questions repeatedly: leave policy, benefits, expense limits. Every question interrupts someone's work."*
>
> *"We track compliance in spreadsheets: work permits, visa renewals, training certifications, equipment sign-offs. It's error-prone and tedious."*
>
> *"Simple requests like updating an address or changing bank details require an HR ticket and back-and-forth emails."*

The current reality: HR spends most of their time on transactional work instead of strategic initiatives. Every new hire generates significant paperwork. Every policy question interrupts someone's day.

Why This Matters Now

HR operations face a fundamental scaling problem:

- Contract generation: Each hire requires multiple customized legal documents
- Policy questions: The same questions get asked repeatedly across the organization
- Compliance tracking: Manual tracking of expirations, certifications, and requirements
- Workflow bottlenecks: Routine requests require human processing when they shouldn't
- Multi-jurisdiction complexity: Different countries have different labor laws and requirements

AI can transform HR from a bottleneck into a self-service function - handling routine operations automatically so HR can focus on people, not paperwork.

The Opportunity

Build an AI-powered self-service platform that makes HR operations invisible:

Phase 1: Intelligent Document Generation:

- Contract Assembly: Takes structured data (role, salary, start date, location) → generates complete employment contract in correct legal format
- Equity Documentation: Auto-generates option grant letters, vesting schedules, exercise agreements
- Localization: Adapts templates for multiple countries with different labor laws
- Version Control: Tracks contract changes, manages amendments, handles renewals

Phase 2: Conversational HR Assistant:

- Policy Questions: Slack/Teams bot answers questions instantly

- "How many days of annual leave do I have?" → pulls from HRIS, shows balance
- "Can I expense coworking space?" → surfaces expense policy with relevant clause
- "When am I eligible for promotion?" → explains criteria, shows progress
- Request Processing: Handles routine changes without HR involvement
- "Update my address" → validates format, updates systems, confirms
- "Add my spouse as a dependent" → collects required info, routes to benefits provider
- Workflow Routing: For complex requests, identifies right approver and routes automatically

Phase 3: Proactive Compliance Intelligence:
- Expiration Tracking: Monitors work permits, visas, certifications, equipment loans
- Well before visa expires → notifies employee and HR with renewal checklist
- Training certification about to lapse → triggers re-certification workflow
- Mandatory Training: Tracks completion, sends reminders, escalates non-compliance
- Audit Readiness: Maintains compliance evidence for labor audits, tax inspections

The system should dramatically reduce time spent on HR operations while improving employee experience.

Constraints

Constraint	Rationale
Must demo live	Show actual contract generation, chatbot interaction, or compliance tracking.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Legal accuracy	Generated contracts must match approved legal templates for each jurisdiction.
Human oversight	AI generates and assists; humans approve legal documents and sensitive changes.
Privacy compliant	Handle employee data responsibly and securely.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Do AI-generated contracts match legal requirements? Does the chatbot understand HR context?
Usefulness	25%	Can the chatbot resolve most HR queries without escalation? Does it save real time?
Craft	20%	Is the employee experience smooth? Would people prefer self-service over waiting for HR?
Ambition	15%	Creative approaches to contract generation, compliance tracking, or workflow automation?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

- How do you ensure AI-generated contracts are legally accurate for each jurisdiction?
- What's the right balance between self-service and human involvement for sensitive HR matters?
- How do you handle ambiguous policy questions where the answer isn't clear-cut?
- Can the system learn from HR responses to improve over time?
- How do you maintain confidentiality when answering employee-specific questions?
- What happens when the bot doesn't know the answer - how does it escalate gracefully?

What Would Blow Our Minds

- Predictive compliance alerts: "Based on hiring plans, you'll need additional visa slots - start applications now"
- Policy gap detection: AI reads employment contracts across multiple countries, identifies inconsistencies, recommends standardization
- Smart contract negotiation: When candidate requests changes, AI pulls market data and generates counter-offer options
- Invisible onboarding: New hire accepts offer → AI automatically generates all contracts, provisions system access, schedules onboarding - zero manual work
- Proactive HR: Bot notices employee hasn't taken leave in months and proactively reminds them of their balance
- Cross-system intelligence: Answers questions by pulling from multiple HR systems and synthesizing a coherent response

Deriv AI Hackathon

Security

Challenge Brief

AI WAF

The Challenge

How might we build an AI WAF (Web Application Firewall) that protects AI agents and LLM applications from prompt injection attacks and malicious inputs?

The Problem

As AI agents become internet-connected and autonomous, they face new attack vectors. Prompt injection can manipulate AI behavior, leak sensitive data, or cause unintended actions. Traditional security tools don't understand LLM-specific attacks.

> *"Our AI agent was manipulated into exposing user data through clever prompt injection. Our WAF didn't catch it."*
>
> *"We have LLMs calling APIs with user input. I have no idea if someone could trick it into doing something malicious."*
>
> *"Every time we patch one prompt injection technique, attackers find new ones."*
>
> *"Traditional input validation doesn't work. The whole point of LLMs is accepting natural language. Where do you draw the line?"*

The core issue: AI systems are fundamentally different from traditional applications. Their flexibility is their strength and their vulnerability. We need new security paradigms for AI.

Why This Matters Now

AI agents are being deployed in production with access to sensitive data and critical actions:

- Prompt injection is the SQL injection of the AI era
- AI agents can take real-world actions - making attacks more dangerous
- Traditional security tools weren't designed for natural language inputs
- New attack techniques emerge faster than manual defenses can adapt
- Organizations need guardrails before widespread AI deployment creates security incidents

AI can be used to detect and defend against AI-specific attacks using adversarial reasoning and intent analysis.

The Opportunity

Build an AI-powered security system that:

- Detects and blocks prompt injection attempts in real-time
- Identifies jailbreak attempts and adversarial inputs
- Validates AI agent outputs before execution
- Monitors AI decision-making for anomalous behavior
- Provides layered defense (input validation, output filtering, behavior monitoring)
- Learns from new attack patterns to improve defenses
- Generates explanations of blocked attacks for security teams
- Provides policy enforcement for AI agent capabilities

The system should protect AI agents without breaking their functionality - stopping attacks while allowing legitimate use.

Constraints

Constraint	Rationale
Must demo live	No slide decks. Show us working software.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Minimize false positives	Blocking legitimate AI use is unacceptable. Security can't break functionality.
Explain blocks	When attacks are blocked, the system must explain why clearly.
Real-time performance	Detection must not add noticeable latency to AI interactions.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	25%	Does your solution understand AI-specific security threats?
Usefulness	30%	Would this actually protect AI systems? Can it stop real attacks?
Craft	20%	Is the UX considered? Is the AI integration thoughtful?
Demo	25%	Can you show real attacks being blocked while legitimate use passes through?

Questions Worth Considering

- What makes a prompt malicious vs. just unusual?
- Can AI detect adversarial intent from natural language alone?
- How do you protect against attacks you haven't seen yet?
- Where should guardrails sit in the AI application stack?
- Can defense mechanisms themselves be vulnerable to prompt injection?
- How do you balance security with AI capabilities?

What Would Blow Our Minds

- A guardrail system that stops prompt injection without blocking legitimate use
- AI that reasons about adversarial intent like a security expert
- Something that adapts to new attack patterns automatically
- Multi-layered defense that validates inputs, outputs, and behavior
- A system that makes AI agents safe to deploy on the public internet
- Defense that explains WHY something was blocked in human-understandable terms

AI Pentester

The Challenge

How might we use AI agents to continuously discover, assess, and prioritize security vulnerabilities across an organization's attack surface?

The Problem

Organizations have sprawling digital infrastructures. New services, APIs, and endpoints are deployed daily. Traditional penetration testing happens quarterly at best. By the time pentesters finish a report, the infrastructure has changed.

> *"We run pentests periodically. Our infrastructure changes daily. The reports are outdated before we finish reading them."*
>
> *"We have hundreds of subdomains and APIs. I don't even know what half of them do or who owns them."*
>
> *"Manual pentesting takes weeks. Finding and exploiting each vulnerability is repetitive detective work."*
>
> *"By the time we fix the vulnerabilities from one pentest, new ones have been introduced."*

The core issue: security testing is event-driven and manual, while modern infrastructure is continuous and dynamic. The attack surface evolves faster than humans can test it.

Why This Matters Now

Modern infrastructure creates new security challenges:

- Cloud-native architectures create ephemeral infrastructure that changes constantly
- APIs multiply as services become more interconnected
- Shadow IT proliferates as teams spin up their own services
- Attack surfaces expand faster than security teams can track
- Periodic testing creates security gaps between assessments

AI agents can automate reconnaissance, vulnerability assessment, and exploit validation - providing continuous security testing that keeps pace with infrastructure changes.

The Opportunity

Build an AI-powered continuous penetration testing system that:

- Automatically discovers and maps the organization's attack surface (domains, subdomains, APIs, services)
- Identifies potential vulnerabilities and misconfigurations using AI reasoning
- Prioritizes findings based on exploitability, impact, and business context
- Generates proof-of-concept exploits to validate vulnerabilities
- Provides actionable remediation guidance in natural language
- Tracks attack surface changes over time and alerts on new exposures
- Generates executive summaries and technical reports automatically

The system should think like a skilled pentester but operate continuously, testing faster and more thoroughly than periodic manual assessments.

Constraints

Constraint	Rationale
Must demo live	No slide decks. Show us working software.
AI must add value	This is an AI hackathon. AI must be core to your solution.
Ethical testing only	All testing must be authorized. No actual attacks on real systems.
Human validation	AI recommends and assists. Critical security decisions require human oversight.
No denial of service	Testing must not disrupt availability of target systems.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	20%	Does your solution understand modern attack surface management challenges?
Usefulness	30%	Would security teams trust this for continuous testing? Does it find real issues?
Craft	15%	Is the UX considered? Is the AI integration thoughtful?
Demo	35%	Working demo of the agent discovering and assessing vulnerabilities.

Questions Worth Considering

- How do you distinguish critical vulnerabilities from noise?
- Can AI understand business context to prioritize risks?
- How do you validate AI-generated exploits safely?
- What's the right balance between automation and human control?
- Can AI learn from past pentests to improve future assessments?
- How do you handle false positives without missing real vulnerabilities?

What Would Blow Our Minds

- A system that discovers zero-day vulnerabilities through creative AI reasoning
- Something that tests many endpoints faster and more thoroughly than a human pentester
- Vulnerability reports that are actually readable and actionable
- A copilot that makes every engineer think like a security expert
- Continuous pentesting that dramatically reduces cost compared to periodic manual testing
- Attack surface mapping that discovers assets the organization didn't know existed

Deriv AI Hackathon

Marketing

Challenge Brief

Autonomous Mystery Shopper

The Challenge

How might we automate the detection of signup friction before a real user ever encounters it? What if your QA team never slept? What if a bot could feel frustration? What if bugs reported themselves to the right person instantly?

The Problem

The mobile/web signup flow is the "front door" for any platform. Every day, potential users click ads and land on your signup page. If the "Sign Up" button hangs for a few seconds, or document upload fails on certain devices, you lose them forever.

Currently, discovering these issues relies on manual QA and in-person product reviews - or worse, waiting for user complaints. By then, the opportunity is gone.

> *"We test what we can, but there are thousands of device/browser/OS combinations. We can't cover them all manually."*
>
> *"A signup bug existed for two weeks before a user complained. How many people did we lose silently?"*
>
> *"When something breaks, we don't know if it's a server error, a UI glitch, or confusing copy. Different problems need different teams."*
>
> *"Our QA team can't test 24/7 across every country and language. Issues slip through."*

The core issue: manual QA doesn't scale to cover every device scenario, network condition, and user journey. By the time bugs are discovered, users are already lost.

Why This Matters Now

With the rise of Autonomous Agents and Multimodal AI, we no longer need humans to manually click through every possible device scenario. We can build intelligent agents that:

- Traverse the app like real humans, using visual cues rather than hard-coded selectors
- Find the cracks in user experience before real users do
- Automatically escalate issues to the right team with full context
- Run continuously, testing different scenarios around the clock

This isn't just a Selenium script that checks if a button exists. This is an AI agent that "looks" at the screen, attempts to sign up, and intelligently reports failures.

The Opportunity

Build an Autonomous "Mystery Shopper" for Mobile & Web UX with three core capabilities:

1. Simulation:
    - AI agent navigates the mobile signup flow purely using visual cues (Computer Vision)
    - Mimics a confused first-time user rather than following a hard-coded script
    - Handles dynamic UI elements that would break traditional automation
    - Tests across different device types, screen sizes, and network conditions

2. Diagnosis:

- When the agent gets stuck, it understands WHY
- Server error (500)? → Backend issue
- UI glitch (button off-screen)? → Frontend/Design issue
- Copy ambiguity ("I don't know what to enter here")? → UX/Content issue
- Timeout or slow response? → Performance issue
- Document upload failure? → Integration issue

3. Escalation:
- The magic happens in the reporting
- Creates a dedicated channel or thread for the issue
- Uploads "evidence" (screenshots, screen recordings, logs)
- Tags the specific human responsible for that part of the journey
- Includes severity assessment and reproduction steps

The system should catch issues that manual QA would miss and route them to the right team instantly.

Constraints

Constraint	Rationale
Must demo live	Show the bot hitting a bug and the alert notification in real-time.
AI must add value	Must use AI (LLM/Vision) to navigate or diagnose. A hard-coded script is not what we're looking for.
Must run mobile	Must simulate a mobile environment (Mobile Web view or Emulator). Desktop flows are not the priority.
Live alerting	Output cannot just be a log file. Must be a live alert (Slack/Teams) with rich media.
Staging environment only	Test on provided staging URL. Do not unleash AI bots on production.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does it catch things a simple script would miss? Can it detect "user confusion" not just code errors?
Usefulness	25%	Is the report immediately actionable? Does it give developers everything needed to fix it?
Craft	20%	How smart is the routing? Does it tag the right department or spam everyone?
Ambition	15%	Did you simulate bad network conditions? Different languages? Use Computer Vision?
Demo	10%	Show us the bot hitting a bug and the notification popping up in real-time.

Questions Worth Considering

- Can the AI determine the "severity" of the bug? (A typo is lower priority, a broken Submit button is critical)
- How does the bot handle CAPTCHA or OTPs?
- Can the bot take a screenshot of the exact moment the error occurred?
- Could this run continuously, testing a different country's flow every hour?
- Can it simulate different user personas (tech-savvy vs. confused first-timer)?
- How does it handle intermittent issues vs. consistent bugs?

What Would Blow Our Minds

- Visual intelligence: Agent navigates purely by "looking" at the screen, not by element IDs
- User confusion detection: "The agent spent excessive time on this screen looking for the next step" → UX issue flagged
- Smart severity scoring: Automatic P0/P1/P2/P3 classification based on impact
- Multi-language testing: Same flow tested in different languages, catching localization bugs
- Network condition simulation: Testing on slow 3G, intermittent connection, high latency
- 24/7 continuous monitoring: Different country/device combinations tested every hour automatically
- Root cause intelligence: "This error is similar to Issue #1234 from last week - possibly related"

Deriv AI Hackathon

Workflow Automation

Challenge Brief

AI-Powered Business Workflow Generator

The Challenge

How might we build an AI system that converts natural language business requirements into fully functional, production-ready automated workflows - without requiring technical expertise?

The Problem

Business operations teams know what they need automated, but building workflows requires technical skills they don't have. They describe processes in plain English, then wait weeks for developers to translate requirements into workflows. Meanwhile, manual work piles up.

> *"I need to automate our employee onboarding process: send welcome email, create accounts in multiple systems, provision access, schedule training. I can explain it in minutes. Building the workflow takes a developer weeks."*
>
> *"We have a multi-step HR onboarding process. Every step involves different systems. Connecting them all manually in a workflow builder takes days of clicking and dragging."*
>
> *"I know which APIs I need to call, but I don't know how to find them in the documentation or how to connect them together."*
>
> *"When workflows break, I can't debug them. I have to wait for a developer to figure out what went wrong."*

The core issue: workflow automation tools require technical expertise. Business teams that understand the processes can't build the automation themselves. The translation layer - from business requirement to working workflow - is a bottleneck.

Why This Matters Now

Every organization has hundreds of repetitive processes that should be automated but aren't:

- Manual work doesn't scale: HR onboarding, expense approvals, data entry, document processing, compliance workflows
- Technical bottleneck: Operations teams depend on developers to build every workflow
- Slow iteration: Changing a workflow requires developer time, creating delays
- Knowledge gap: Business experts know the process, developers know the tools - neither has both

Low-code workflow tools help, but still require:
- Understanding API documentation
- Configuring authentication and credentials
- Mapping data between systems
- Debugging integration errors
- Testing and validation

Generative AI can bridge this gap - turning conversational requirements into executable workflows.

The Opportunity

Build an AI-powered workflow generator that:

1. Conversational Requirement Collection:
   - Natural language input: Business users describe what they need in plain English
   - Example: "When a new employee is hired, send them a welcome email, create their accounts, add them to Slack, and schedule their first-week training"
   - Intelligent clarification: AI asks targeted questions to gather missing information
   - Process decomposition: Breaks complex requirements into logical workflow steps

2. Intelligent API Discovery & Mapping:
   - Semantic API search: Finds the right APIs from documentation based on what you're trying to do
   - Multi-system integration: Connects APIs across different platforms (CRM, HR systems, email, databases, cloud services)
   - Authentication handling: Identifies required credentials and guides configuration
   - Data mapping: Figures out how to pass output from one step as input to the next

3. Automated Workflow Generation:
   - Workflow builder: Generates executable workflows in target platform format
   - Error handling: Builds in retry logic, failure notifications, fallback paths
   - Testing framework: Automatically generates test cases to validate the workflow
   - Iterative fixing: If workflow fails during testing, AI debugs and fixes automatically

4. Human-in-the-Loop Refinement:
   - Visual workflow preview: Shows what the AI built before deploying
   - Modification support: User can request changes in natural language ("Add a delay before sending the reminder")
   - Approval gates: Business owner approves before production deployment
   - Continuous improvement: Learns from feedback to improve future workflows

The system should turn a short conversation into a working, tested workflow - dramatically reducing the time from requirement to deployment.

Constraints

Constraint	Rationale
Must demo live	Show actual workflow generation from conversation.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Non-technical friendly	Business users should understand the conversation without technical jargon.
Executable output	Generated workflows must actually work, not just look right.
Human oversight	AI generates, humans approve deployment.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Can non-technical users describe requirements successfully?
Usefulness	25%	Does it find the right APIs from natural language descriptions?
Craft	20%	Do generated workflows actually execute successfully?
Ambition	15%	Creative approaches to requirement interpretation or error handling?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

- How do you handle ambiguous requirements without asking too many clarifying questions?
- What's the right balance between AI autonomy vs. user control?
- How do you discover which APIs exist when users don't know system names?
- Can you generate workflows that work across different workflow platforms?
- How do you handle authentication/credentials securely?
- What happens when a required API doesn't exist - can AI suggest alternatives?

What Would Blow Our Minds

- End-to-end generation: "I need to automate employee onboarding" → short conversation → working multi-step workflow deployed
- Intelligent API discovery: User describes intent in plain language → AI finds correct endpoint from many APIs across multiple systems
- Self-debugging: Workflow fails during test → AI reads error logs → fixes the workflow → re-tests automatically
- Multi-platform support: Same conversation generates workflows for different platforms depending on user preference
- Learning from examples: Analyzes existing workflows to understand patterns and improve future generation
- Natural modifications: "Actually, send that email later instead" → workflow updated instantly without rebuilding

Deriv AI Hackathon

Data

Challenge Brief

Real-Time Enterprise Intelligence Agent

The Challenge

How might we build an AI-powered enterprise intelligence platform that unifies data across all business domains - clients, finance, partners, operations, compliance, and competition - to surface real-time insights and drive informed decision-making?

The Problem

Enterprise data lives in silos. Client data sits in CRM. Financial data lives in ERP. Partner performance is tracked separately. Competitive intelligence comes from yet another source. When executives need to make decisions, they're piecing together information from multiple systems, multiple reports, and multiple teams - often with data that's hours or days old.

> *"I need to understand why client acquisition costs are rising. Is it a marketing problem? A competitive problem? A product problem? A regional problem? I have to pull data from five different systems and spend days triangulating."*
>
> *"Our finance team sees revenue trends. Our operations team sees conversion rates. Our partner team sees referral quality. Nobody connects the dots across all three to see the full picture."*
>
> *"By the time we compile our weekly executive report, the data is already stale. We're making decisions based on what happened last week, not what's happening now."*
>
> *"A competitor dropped their pricing. How does that affect our client acquisition? Our partner economics? Our revenue forecast? It takes weeks to model the impact across all domains."*
>
> *"We have all the data. We just can't see it together, in real-time, with AI helping us understand what it means."*

The core issue: enterprises are data-rich but insight-poor. Data exists across every domain, but connecting it, analyzing it in real-time, and translating it into actionable decisions requires massive manual effort.

Why This Matters Now

Modern enterprises generate vast amounts of data across every function, but extracting value requires connecting the dots:

- Client Intelligence: Acquisition costs, conversion rates, lifetime value, churn signals, behavioral patterns
- Financial Intelligence: Revenue trends, cost structures, margin analysis, cash flow, forecast accuracy
- Partner Intelligence: Referral quality, partner performance, commission economics, relationship health
- Operational Intelligence: Platform health, transaction success rates, system performance, capacity utilization
- Competitive Intelligence: Market positioning, pricing movements, feature comparisons, win/loss analysis
- Compliance Intelligence: Regulatory exposure, risk concentrations, audit readiness, policy adherence

Current challenges:
- Data silos: Each domain has its own systems, reports, and teams
- Delayed insights: Reports are generated daily or weekly, not in real-time
- Manual correlation: Connecting insights across domains requires analyst effort
- Missing context: Financial metrics without client context, or client metrics without competitive context
- Reactive decisions: Problems discovered after impact, not predicted in advance

AI can unify these domains, surface cross-functional insights in real-time, and recommend actions based on the complete picture.

The Opportunity

Build an AI-powered enterprise intelligence platform with five core capabilities:

1. Unified Data Layer & Real-Time Integration:
   - Cross-domain data ingestion: Connect client, financial, partner, operational, and competitive data sources
   - Real-time streaming: Process data as it arrives, not in daily batches
   - Entity resolution: Link the same client, partner, or transaction across different systems
   - Data quality monitoring: Flag inconsistencies, gaps, and anomalies in source data
   - Historical context: Maintain time-series data for trend analysis and pattern recognition

2. Cross-Domain Intelligence & Pattern Recognition:
   - Multi-domain correlation: Automatically identify relationships across business functions
   - Example: "Client acquisition costs rose because competitor X dropped pricing in Region Y, causing our partner referral quality to decline"
   - Example: "Revenue forecast accuracy improved in markets where we increased operational capacity"
   - Anomaly detection: Surface unusual patterns that span multiple domains
   - Trend identification: Recognize emerging patterns before they become obvious
   - Causal analysis: Distinguish correlation from causation using AI reasoning

3. Predictive Analytics & Scenario Modeling:
   - Forecasting: Project client growth, revenue, costs, and operational needs
   - Scenario simulation: "What happens to our P&L if competitor X acquires competitor Y?"
   - Risk modeling: Identify concentration risks across clients, partners, regions, or products
   - Sensitivity analysis: Which variables most impact key business outcomes?
   - Early warning: Predict problems before they materialize based on leading indicators

4. Natural Language Interface & Conversational Analytics:
   - Ask anything: Executives query the platform in natural language
   - "Why is client lifetime value declining in APAC?"
   - "How does our partner mix compare to last quarter?"
   - "What's driving the variance in our revenue forecast?"
   - "If we increase marketing spend, what's the projected ROI?"
   - Follow-up questions: Drill down through conversation, not clicks
   - Proactive insights: Platform surfaces important changes without being asked

5. Decision Support & Action Recommendations:
   - Actionable recommendations: Not just "what happened" but "what to do about it"

- Example: "Client acquisition costs are rising due to competitive pressure in Region Y. Recommend: (1) Adjust partner incentives, (2) Launch retention campaign for at-risk clients, (3) Review pricing in affected segments"
- Impact quantification: "This recommendation would improve margin by X% and reduce churn by Y%"
- Trade-off analysis: Present options with pros, cons, and confidence levels
- Decision tracking: Monitor outcomes of decisions to improve future recommendations

The platform should feel like having a Chief Intelligence Officer who sees everything, connects everything, and advises on everything - in real-time.

Data Domains & Example Use Cases

Client Intelligence:
- Real-time acquisition funnel: Traffic → Leads → Conversions → Active Clients
- Cohort analysis: How do clients acquired this month compare to previous cohorts?
- Churn prediction: Which clients are at risk and why?
- Lifetime value optimization: What drives higher LTV clients?

Financial Intelligence:
- Revenue decomposition: By product, region, client segment, partner channel
- Cost analysis: Marketing efficiency, operational costs, partner commissions
- Margin optimization: Where are we making money? Where are we losing it?
- Forecast accuracy: How well did our predictions match reality?

Partner Intelligence:
- Partner performance: Traffic quality, conversion rates, client retention
- Commission economics: ROI by partner tier, channel, region
- Relationship health: Engagement trends, satisfaction signals, churn risk
- Optimization opportunities: Which partners should we invest in?

Competitive Intelligence:
- Market positioning: Where do we win? Where do we lose?
- Pricing dynamics: How do competitor moves affect our performance?
- Feature comparison: Gap analysis against market
- Trend monitoring: What are competitors doing that we should know about?

Operational Intelligence:
- Platform health: Success rates, error rates, latency, capacity
- Incident correlation: How do operational issues impact business metrics?
- Capacity planning: Do we have headroom for growth?
- Efficiency optimization: Where can we reduce costs without impacting quality?

Constraints

Constraint	Rationale
Must demo live	Show real-time data analysis and AI-driven insights in action.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Cross-domain integration	Must connect insights across multiple business domains, not just one silo.
Real-time processing	Analysis and recommendations must reflect current state, not yesterday's data.
Actionable output	Insights must drive decisions, not just inform. Recommend specific actions.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does it surface cross-domain insights humans would miss? Are connections meaningful?
Usefulness	25%	Would executives and teams use this for real decisions? Does it save analysis time?
Craft	20%	Is the interface intuitive? Are recommendations clear and well-reasoned?
Ambition	15%	How many domains connected? Predictive capabilities? Scenario modeling?
Demo	10%	Can you show end-to-end: data → insight → recommendation → decision support?

Questions Worth Considering

- How do you handle data quality issues across different source systems?
- What makes a cross-domain insight valuable vs. just interesting?
- How do you present complex multi-factor analysis in a digestible way?
- When should the platform proactively surface insights vs. wait to be asked?
- How do you balance depth of analysis with speed of response?
- Can you explain AI reasoning in terms executives understand?
- How do you handle uncertainty and confidence levels in recommendations?
- What's the right interface for different users (CEO vs. analyst vs. operator)?

What Would Blow Our Minds

- True cross-domain reasoning: "Client LTV is declining because partner mix shifted toward lower-quality channels, which happened because our commission structure became uncompetitive after competitor X raised partner payouts"
- Predictive early warning: "Based on current trends, we'll miss Q4 revenue target by X%. Root causes: competitive pressure (40%), partner churn (35%), operational issues (25%). Recommended interventions prioritized by impact."
- Scenario simulation: "If competitor X acquires competitor Y, model shows: client acquisition costs +15%, partner defection risk for top 20 partners, recommended defensive actions with ROI estimates"
- Real-time executive briefing: "Good morning. Here's what changed overnight: 3 insights requiring attention, 2 emerging risks, 1 opportunity identified. Shall I elaborate on any?"
- Natural conversation: Executives can ask follow-up questions, request drill-downs, challenge assumptions, and explore alternatives through dialogue
- Learning from decisions: Platform tracks which recommendations were implemented and their outcomes, improving future suggestions
- Unified view: Single source of truth that eliminates the need to reconcile multiple reports and dashboards
- Something that makes executives say: "This is like having an entire analytics team available 24/7 who actually understands our business."